Twelve trading ideas, tested honestly on free data

A pre-registered negative-results report on short-term US stock strategies

6 November 2026

A personal research project. Not investment advice; it recommends no security or strategy. No real money was traded. The code, analysis and drafting were done with an AI coding assistant; every number comes from results recorded in the project repository.

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Abstract

I set out to build a pipeline that could find a short-term trading edge in US stocks starting from zero capital, and to trust it only if it survived hard tests. Over ten test milestones I tested twelve strategy ideas, from opening-range breakouts and RSI reversion to earnings drift, insider buying, volume signals, indicator bots popular in online videos, presidential social-media posts and Pentagon contract awards. Every test was written down and committed before it was run, scored out of sample, and charged for the number of tries it took to find it.

None of the twelve ideas met all of its pre-registered criteria. Several looked promising until the market's own rise was subtracted or until the idea was repeated on stocks it had never seen. No money was traded, and no idea was promoted to paper trading for evidence. The useful output is not a strategy; it is a method, and a short list of ways that backtests flatter themselves.

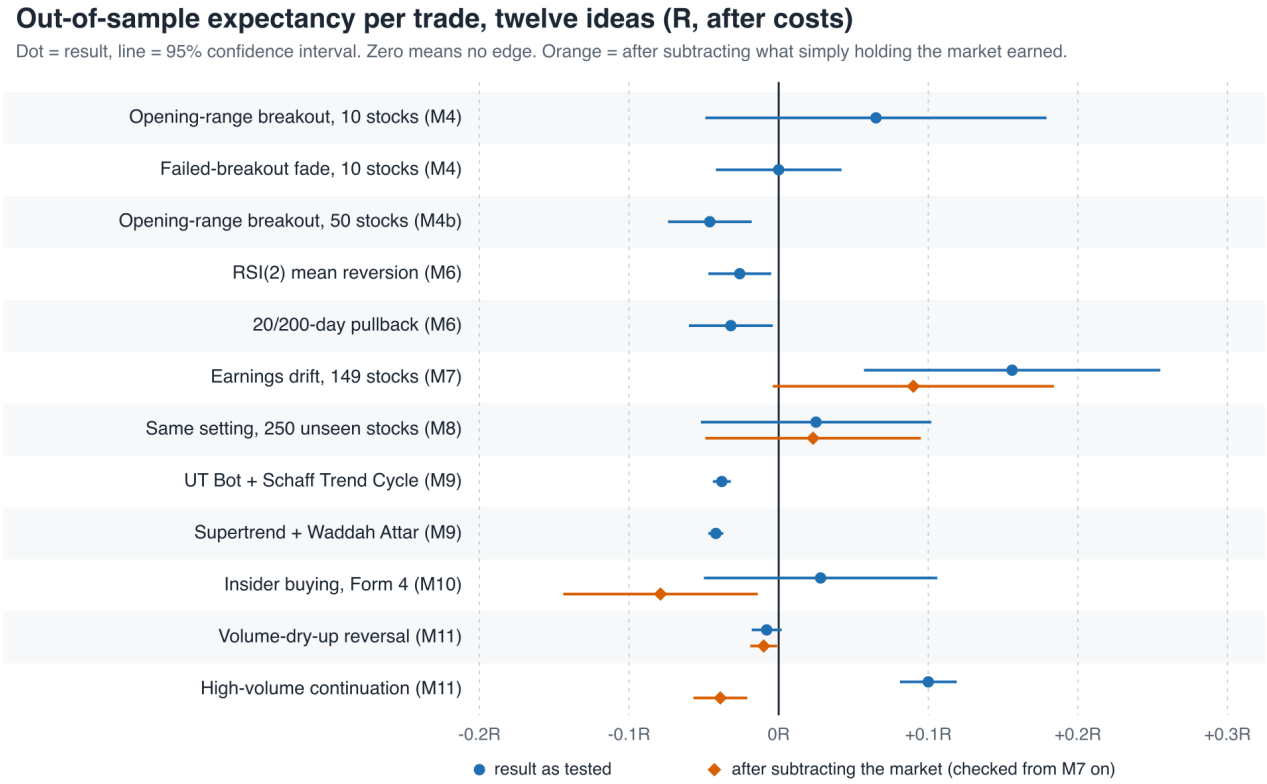


Figure 1. Out-of-sample result per trade, in R (profit measured in units of the amount risked). Zero means no edge. Blue: as tested. Orange: after subtracting what simply holding the market earned over the same dates, a check added from M7 on. Lines are 95% confidence intervals.

Summary

- **No idea passed.** Of twelve, the closest were earnings drift (5 of 6 checks, then it failed to replicate on 250 unseen stocks) and a volume-spike rule (5 of 7 checks, then it turned negative after subtracting the market).
- **When it looked like an edge, it was mostly the market.** Long-only rules were positive in 2019 to 2024 largely because stocks rose. Insider buying went from +0.028R to −0.079R, and volume spikes from +0.100R to −0.039R, once the market was subtracted.
- **Costs were not the main reason.** The price-pattern families had no edge even at zero cost (for example breakouts at −0.020R, volume reversal at +0.006R, both consistent with zero).
- **Replication was the strongest filter.** A faint breakout result on 10 stocks (+0.065R) became −0.046R on 50 fresh ones. Earnings drift (+0.156R) became +0.025R on 250 unseen stocks.
- **Precision was high, so the negatives are real.** Some tests had around 100,000 trades with intervals of about ±0.005R. These are measured zeros and small negatives, not "not enough data".
- **A side question gave a trade-off, not a free lunch.** A 200-day trend filter on SPY cut the worst drawdown from −32% to −22% but cost about 4 points a year of return.

1. Question and constraints

The goal: a pipeline that gathers data from trusted free sources, applies rules, backtests and optimises them, and only then paper trades, with real money considered only after a proven edge. Constraints fixed at the start:

- US stocks, short-term horizons (minutes to a few weeks), a fixed 1:1.5 risk-to-reward bracket on trades.
- Start with $0 of capital. Free data only: Alpaca's free tier (consolidated-tape bars from 2016, 15 minutes delayed), SEC EDGAR filings, USAspending.gov, public archives of presidential posts.
- Universe chosen by a fixed rule (median dollar volume in Q1 2016), never by hand: 10 large caps, then 50, 150 and 250 further stocks (about 410 in total).

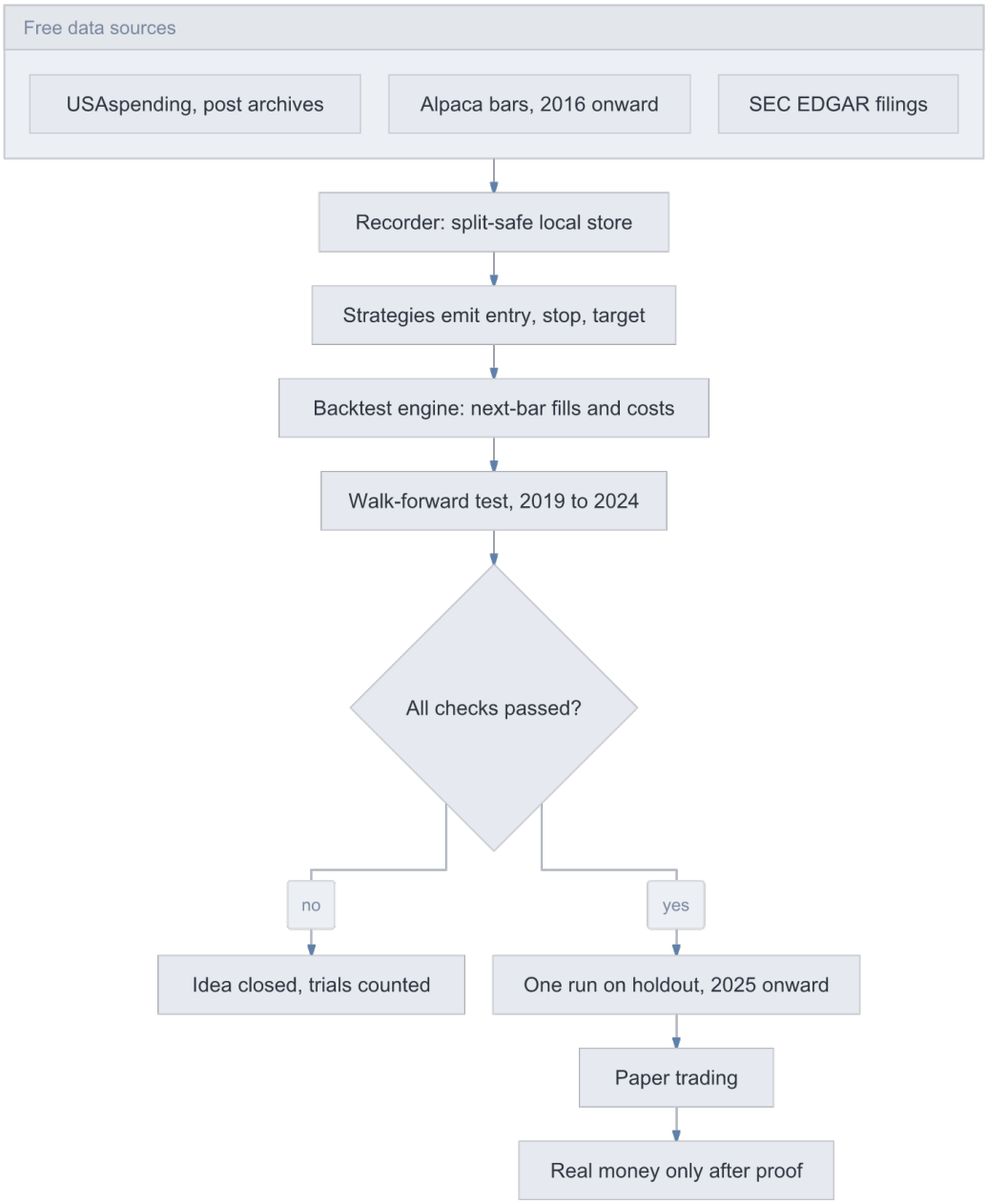


Figure 2. The pipeline. Data flows from free sources through a split-safe store into strategies and a cost-aware backtest; only an idea that passes every check reaches the holdout, and only after that paper trading.

2. How the tests were kept honest

Most backtests that look good are noise that was selected. The framework was built to make that hard.

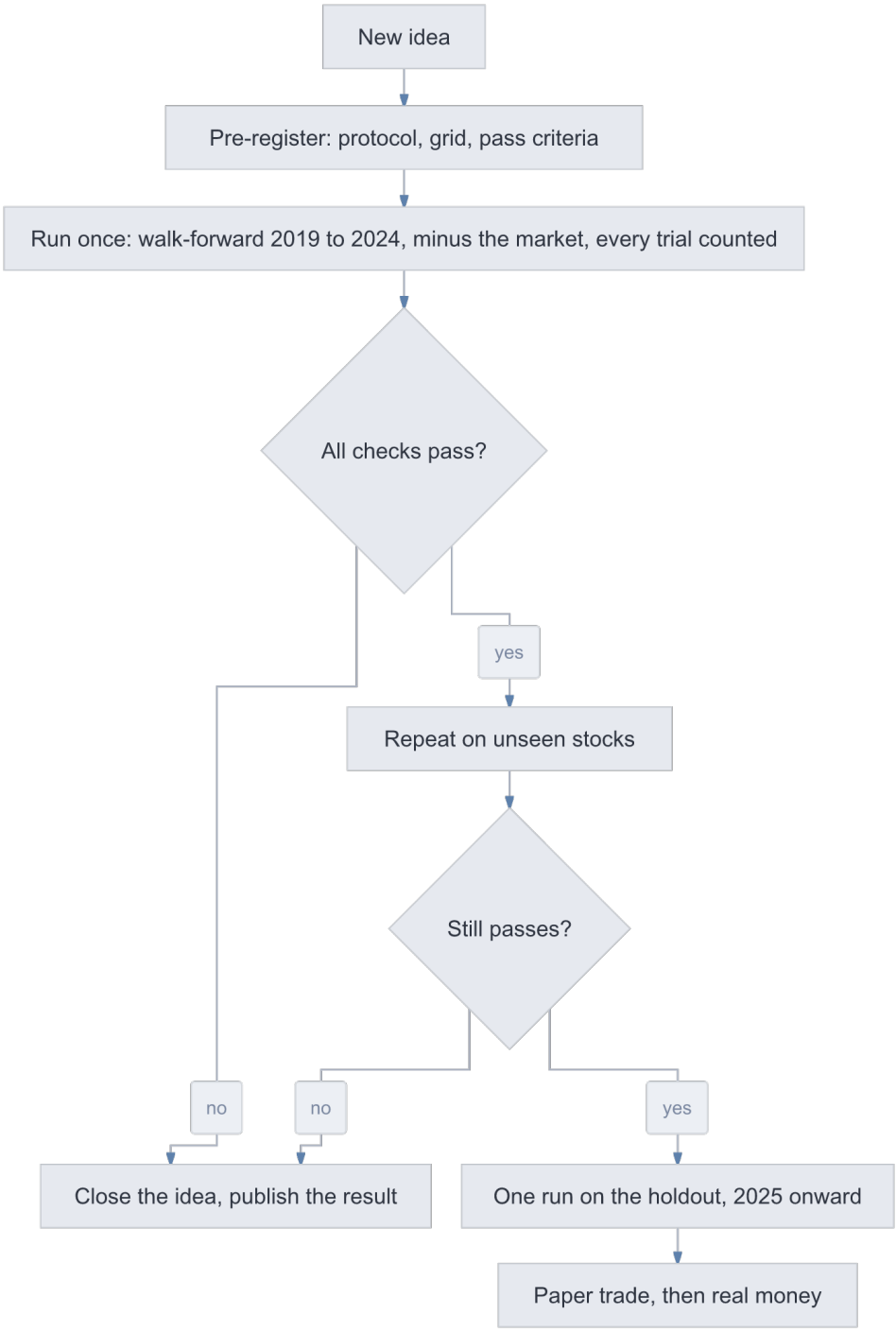


Figure 3. The method applied to every idea. Failing at any gate closes the idea and the result is published either way.

Safeguard	What it does
Pre-registration	For each test the protocol, parameter grid and pass criteria were committed to version control before the run. The commit history shows the order, so it cannot be quietly changed afterwards.
Walk-forward, out of sample	Parameters are chosen on earlier years only and judged on the next. Test years were 2019 to 2024.

Safeguard	What it does
Untouched holdout	Data from 2025 onward is kept for a single final run, and only for an idea that has already passed everything. The code refuses to read it without an explicit flag. No analysis has read it.
Trial counting and Deflated Sharpe	Every configuration tried is logged (298 in total). The Deflated Sharpe Ratio (Bailey and Lopez de Prado) asks whether a result still stands after paying for the number of attempts.
Realistic fills and costs	Next-bar fills, stops assumed hit first when a bar is ambiguous, gaps filled at the open, per-share slippage on market fills, limit targets that must trade through to fill, stock splits handled at read time, one open trade per symbol. A stress test doubles costs.
Market-adjusted check	Added after earnings drift: each trade's result minus what holding SPY the same way over the same dates would have earned.
Replication on unseen stocks	A single frozen setting is run once on stocks that took no part in choosing it.

Pass criteria (all must hold): at least 300 out-of-sample trades; the 95% interval of expectancy above zero; Deflated Sharpe at least 0.95; positive in at least 4 of 6 test years; still positive at double costs; at least 70% of neighbouring parameter settings positive; and, from M8 on, a market-adjusted interval above zero. A test with too few trades is "inconclusive", not a pass.

A note on units: **R** is profit measured in the amount risked on the trade. +0.1R means the average trade made 10% of what it risked, after costs.

3. Results

Milestone	Idea	Data	Out-of-sample result (95% interval)	Checks passed
M4	Opening-range breakout	10 large caps, 5-min bars	+0.065R ±0.114	3 of 6
M4	Failed-breakout fade	same	−0.000R ±0.042	1 of 6
M4b	Opening-range breakout	50 fresh stocks	−0.046R ±0.028	1 of 6
M6	RSI(2) mean reversion	50 stocks, daily	−0.026R ±0.021	1 of 6
M6	20/200-day pullback	50 stocks, daily	−0.032R ±0.028	1 of 6
M7	Earnings drift (SEC filing times)	149 stocks, daily	+0.156R ±0.099	5 of 6
M8	M7's frozen setting, replicated	250 unseen stocks	+0.025R ±0.077	2 of 7
M9	UT Bot + Schaff Trend Cycle	50 stocks, 5-min	−0.038R ±0.006	1 of 6
M9	Supertrend + Waddah Attar	50 stocks, 5-min	−0.042R ±0.005	1 of 6
M10	Insider buying (SEC Form 4)	407 stocks, daily	+0.028R ±0.078	3 of 7
M11	Volume-dry-up reversal	409 stocks, daily	−0.008R ±0.010	1 of 7
M11	High-volume continuation	409 stocks, daily	+0.100R ±0.019	5 of 7
M12	Presidential posts naming a listed company	133 events	−0.05% ±0.60 per event	1 of 4
M12	Large Pentagon awards to listed primes	933 events	−0.10% ±0.27 per event	1 of 4

M12 is an event study (average market-adjusted return over five sessions, net of 10 basis points), not a strategy sweep, so it has its own four criteria. "M4" and "M4b" are the ten-stock and fifty-stock runs of the same breakout grid.

Details, protocols and per-year numbers for each row are in `docs/m*-protocol.md` and `docs/m*-results.md`.

4. What kept recurring

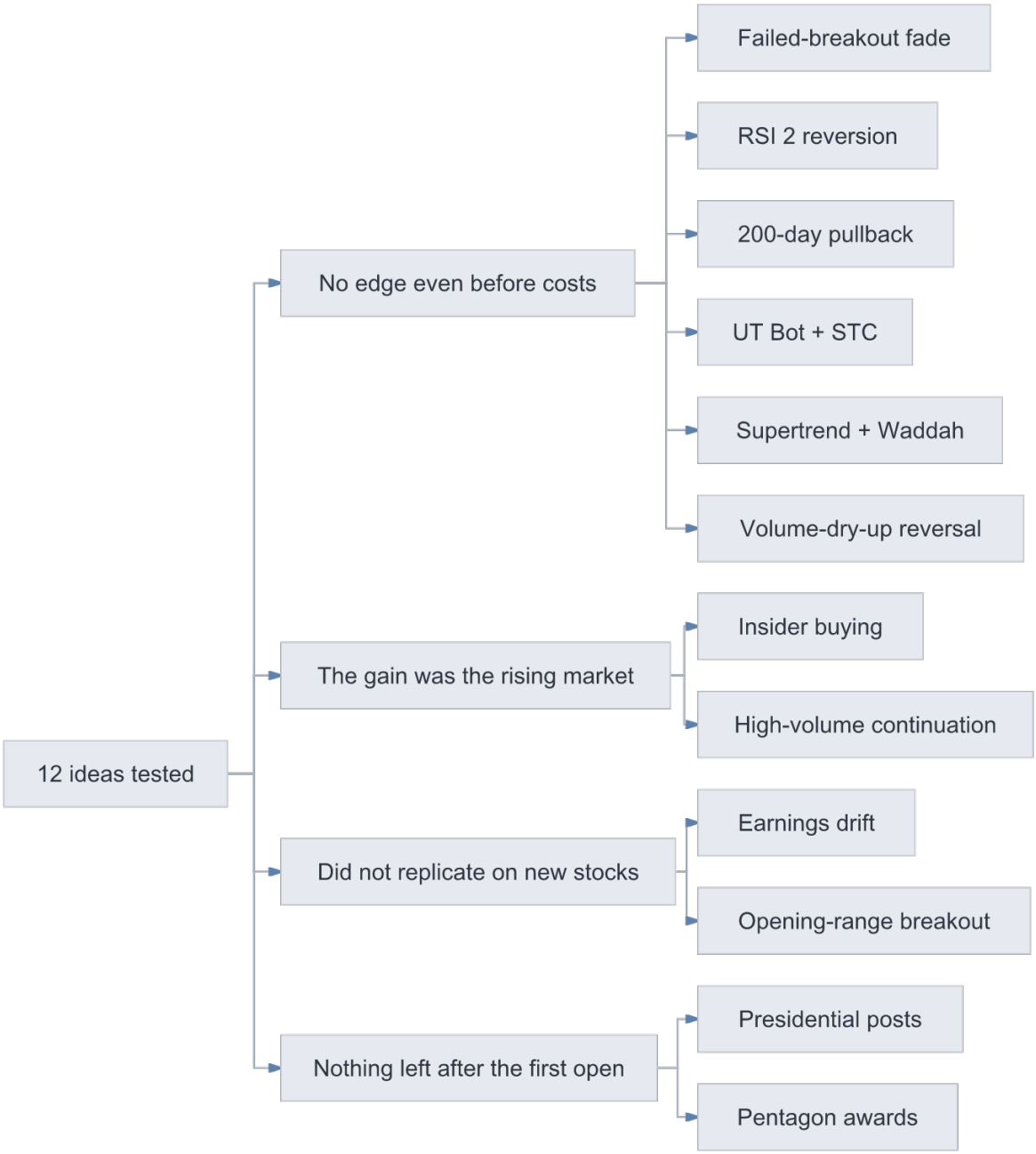


Figure 4. Why each idea failed, by main reason. Several had more than one: the breakout, for example, also showed a faint edge on 10 stocks that vanished on 50.

Costs were not the explanation. The first guess was that a one-cent spread against tight five-minute stops ate a small edge. A zero-cost diagnostic showed otherwise: breakouts were −0.020R ±0.028 before costs and volume reversal +0.006R ±0.010. There was no gross edge to eat, so cheaper execution would not have rescued these ideas.

The market did the work. The long-only families look strongest in 2019 to 2024, a period in which US stocks rose. The market-adjusted check changed the picture:

Idea	As tested	After subtracting the market
Earnings drift (M7)	+0.156R	+0.090R ±0.094
Insider buying (M10)	+0.028R	−0.079R ±0.065
High-volume continuation (M11)	+0.100R	−0.039R ±0.018

All 16 volume-spike configurations were positive as tested, and they still failed. A related effect: the stock universe is made of companies that were liquid in 2016 and are still listed today. Survivors make long positions look better and short positions worse (in M6 longs were near zero and shorts clearly negative).

Lucky results did not replicate. The ten-stock breakout test (+0.065R) turned into −0.046R on fifty fresh stocks, with all 54 configurations negative. Earnings drift passed five of six checks and then fell to +0.025R on 250 unseen stocks. The Deflated Sharpe check had flagged the first of these in advance; replication confirmed it.

Several attractive stories went nowhere. Distinguishing routine from opportunistic insider trades follows a well-known paper (Cohen, Malloy and Pomorski, 2012); in our large and mid-cap universe there were too few opportunistic buys to test the split. Presidential posts: published work finds same-day reactions to tweets about firms (Ajjoub, Walker and Zhao), but starting at the next open, as a slower trader must, nothing was left.

5. A side question: does a trend filter make holding the market safer? (M13)

Since the market kept beating the rules, I tested the simplest allocation rule: hold SPY while it is above its 200-day average, otherwise hold cash. One parameter, fixed in advance, 2017 to 2024:

	Buy and hold	200-day filter
Return per year	12.8%	8.8%
Worst drawdown	−32.4%	−21.8%
Sharpe (risk-adjusted, no cash yield)	0.78	0.75

It reduced the worst falls (2020: −19% instead of −32%; 2022: −17% instead of −27%) but missed the rebounds, and 18 of its 26 in-and-out trades lost money. It passed two of three pre-registered checks and failed the return one by about 4 points a year. Eight years and two crashes are a short sample, and this is research, not advice: how large a fall a holder can sit through matters more than these figures.

6. What this does not show

Absence of evidence in these tests is not proof that no edge exists. The scope was narrow:

- **Free data and daily or five-minute bars.** I could not test the first seconds and minutes after news, where fast traders operate, nor order-book or options data, nor paid datasets (earnings estimates, short interest, intraday news).
- **Large and mid caps only.** Several published effects are strongest in small stocks. I could not test them honestly because free price history omits delisted companies, which biases any such result upward.
- **One regime.** 2016 to 2024 is one stretch of one market, dominated by a rise. The holdout (2025 onward) was kept unread for a final run that no idea earned.
- **Survivor and hindsight bias.** The universe is stocks listed today. This was disclosed in each protocol and is why the market-adjusted check matters.

- **Clustering.** Events bunch in market selloffs, so confidence intervals are somewhat optimistic about independence.
- **Backtests are not live trading.** Slippage in fast markets, rejected orders and partial fills are modelled approximately. A paper-trading loop was built and tested against a broker's paper account for plumbing only; nothing was run for evidence.

7. Takeaways

1. **Write the rules down before you look.** Commit them. It removes the most common way to fool yourself.
2. **Count your tries.** The best of many looks good by luck; deflate for it.
3. **Subtract the market.** In a rising period a long-only rule needs to beat holding, not zero.
4. **Replicate on stocks that took no part in choosing the rule.** It was the strongest filter here.
5. **Ask whether there is a gross edge before blaming costs.**
6. **Treat screenshots, reels and "I made X overnight" posts as hypotheses.** They are one run, picked after the fact, and rarely come with losing runs or costs. They deserve the same test as anything else, and most will fail it.
7. **A clear "no" is a result.** It saved capital, and the method now makes the next question cheap to test.

8. Reproduce it

Everything is in the project repository: code, each protocol (committed before its run), each result write-up and the tests (204 passing). Data is not redistributed; the scripts fetch it from the free sources above.

```
uv sync
uv run pytest
uv run python scripts/sweep.py --config config/sweeps/m10-insider.toml
```

The last command reruns one pre-registered sweep (insider buying).

Start with `README.md`, then `docs/m4-protocol.md` for the method and `docs/m*-results.md` for each result.

9. Disclosures

This is a personal research project. It is not investment advice and recommends no security or strategy. No real money was traded. The code, analysis and drafting were done with an AI coding assistant (Claude Code); every number in this report is taken from results recorded in the repository, and the protocols' commit order is the evidence of what was decided when.

References

- Bailey, D. and Lopez de Prado, M. (2014), The Deflated Sharpe Ratio.
- Cohen, L., Malloy, C. and Pomorski, L. (2012), Decoding Inside Information, Journal of Finance 67(3), record.
- Ajjoub, C., Walker, T. and Zhao, Y., Social media posts and stock returns: The Trump factor, International Journal of Managerial Finance, record.
- Martineau, C. (2022), Rest in Peace Post-Earnings Announcement Drift, Critical Finance Review 11(3-4), record: argues the drift has largely disappeared outside the smallest stocks, which is why earnings drift had a low prior here. Later work disputes this, depending on whether micro-caps are included.